***p. 198** … it can be shown that really there would be no oscillation. The system would converge to a stable state.*

I am sorry to say that this statement is wrong. It is wrong in an interesting way, however, so I have left the error in and shall now take some time to expose it. It is actually the same kind of error as Gale and Eaves spotted in Maynard Smith and Price's original paper (see note to page 97). My error was pointed out by two mathematical biologists working in Austria, P. Schuster and K. Sigmund.

I had correctly worked out the ratios of faithful to philanderer males, and of coy to fast females, at which the two kinds of males were equally successful, and the two kinds of females were equally successful. This is indeed an equilibrium, but I failed to check whether it was a *stable* equilibrium. It could have been a precarious knife-edge rather than a secure valley. In order to check for stability, we have to see what would happen if we perturb the equilibrium slightly (push a ball off a knife-edge and you lose it; push it away from the centre of a valley and it comes back). In my particular numerical example, the equilibrium ratio for males was faithful and philanderer. Now, what if by chance the proportion of philanderers in the population increases to a value slightly higher than equilibrium? In order for the equilibrium to qualify as stable and self-correcting, it is necessary that philanderers should immediately start doing slightly less well. Unfortunately, as Schuster and Sigmund showed, this is not what happens. On the contrary, philanderers start doing better! Their frequency in the population, then, far from being self-stabilizing, is self-enhancing. It increases—not for ever, but only up to a point. If you simulate the model dynamically on a computer, as I have now done, you get an endlessly repeating cycle. Ironically, this is precisely the cycle that I described hypothetically on pages 197–8, but I thought that I was doing it purely as an explanatory device, just as I had with hawks and doves. By analogy with hawks and doves I assumed, quite wrongly, that the cycle was hypothetical only, and that the system would really settle into a stable equilibrium. Schuster and Sigmund's parting-shot leaves no more to be said:

> Briefly, then, we can draw two conclusions:
>
> (a) that the battle of sexes has much in common with predation; and
> (b) that the behaviour of lovers is oscillating like the moon, and unpredictable as the weather.

Of course, people didn't need differential equations to notice this before.

***p. 202** … cases of paternal devotion … common among fish. Why?*

Tamsin Carlisle's undergraduate hypothesis about fish has now been tested comparatively by Mark Ridley, in the course of an exhaustive review of paternal care in the entire animal kingdom. His paper is an astonishing *tour de force* which, like Carlisle's hypothesis itself, also began as an undergraduate essay written for me. Unfortunately, he did not find in favour of the hypothesis.

***p. 206** … a kind of unstable, runaway process.*

R. A. Fisher's runaway theory of sexual selection, which he stated extremely briefly, has now been spelt out mathematically by R. Lande and others. It has become a difficult subject, but it can be explained in nonmathematical terms provided sufficient space is given over to it. It does need a whole chapter, however, and I devoted one to it in *The Blind Watchmaker* (chapter 8), so I'll say no more about it here.

Instead, I'll deal with one problem about sexual selection that I have never sufficiently emphasized in any of my books. How is the necessary variation maintained? Darwinian selection